Thomas Dybdahl Ahle

Copenhagen, Denmark

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October 2026

I'm a computer scientist with expertise in algorithms, machine learning and software engineering. I did my PhD in Locality Sensitive Hashing in Copenhagen (2019), while also co-founding an award-winning NLP startup, 2018-19, as Chief of ML. As Head of Machine Learning at Normal Computing, I build AI systems for chip design and formal verification. Before that, as a Research Scientist at Meta, I worked on scaling language models and ranking.

Education

IT University of Copenhagen

- Ph.D. in Computer Science, June 2019.
- Advisors: Rasmus Pagh, Eric Price.
- Thesis: *Algorithms for Sets and other High Dimensional Data.*

IT University of Copenhagen & University of Copenhagen

- Master of Science in Computer Science, 2017.
- Advisors: Rasmus Pagh, Mikkel Thorup.
- Thesis: *Minhash without false negatives.*

University of Oxford (top 1st)

- Bachelor of Arts in Computer Science, 2013.
- Master of Arts in Computer Science, 2019.
- Advisors: Oege de Moor, Samson Abramsky.
- Thesis: *The Donkey and the Monad.*

Industry

Head of Machine Learning at Normal Computing, 2023 - Present.
I developed the machine learning strategy for Normal Computing, which had previously been focused on blue-sky research in thermodynamic computing. I built the team from scratch, hiring 10 engineers and researchers from top companies and universities. We developed an AI system for the US's largest chip manufacturer, which allowed creating formal models for new chips, which previously took months, in just a few hours. While at Normal Computing I published five papers on algorithms for thermodynamic computing, as well as one about a new approach to long-context language models.

Board of Directors at Silicon Integration Initiative (Si2), 2026 - 2027.
I was elected to the board of Si2, a not-for-profit consortium of about 70 semiconductor companies that develops shared standards for chip design.

Research Scientist at Meta, 2020 - 2023.
I led and co-founded the Machine Learning Efficiency group, a five-person internal applied research group focused on scaling AI across the organisation. During the first year, we developed a new hashing-based algorithm that reduced the size of the internal recommendation systems by 50%. In another project, we devised a new inference algorithm for Bayesian Neural Networks, allowing the Integrity team to deploy well-calibrated models directly on customer devices. Both projects also resulted in publications at top-tier conferences. Finally, we worked on scaling transformer models, lowering the memory requirement and inference time of Facebook Assistant through the use of embedding table compression, smart attention, and a variety of other tricks.

Chief Machine Learning Officer at SupWiz, 2017 - 2018.
I co-founded an NLP start-up with University of Copenhagen academics. At SupWiz, I led a four-person team in developing our chatbot software and putting it into production at three of Denmark's largest IT firms. (There are now many more.) We used a combination of traditional symbolic AI and modern (at the time) sentence embeddings. The chatbot was awarded the most prestigious prize by the Innovation Fund Denmark in 2019. I was also in charge of our hiring efforts, interviewing dozens of candidates and hiring four engineers over the course of five months.

Software Engineer at Other: Sophion, Palantir and XION, 2010 - 2014.

Through various software engineering jobs, I have gained broad exposure to the different areas of software development. At Sophion Bioscience, I developed internal debugging tools for sifting through gigabytes of datasecond on ion channel screening machines. At Palantir, I ported the Metropolis ontological time-series system (now Foundry) to the web, designing data visualization and efficient processing pipelines. At XION, I Developed the most popular Danish TV listings app for Android at the time, based on data scraped (consensually) from hundreds of TV-station websites.

Awards and Scholarships

Research Travel Award, Stibo-Foundation, 2016.
Highly competitive scholarship, given to just two Danish students a year to collaborate in research abroad.

Northwestern Europe Regional Programming Contest, 1st, Association for Computing Machinery, 2014.
As a member of Lambdabamserne, my team and I made history by becoming the first ever Danish team to qualify for the ACM world finals after winning 1st place out of all universities in North Western Europe.

Danish National Programming Champion, 1st, Netcompany, 2013, 2014.
Crowned Danish National Programming Champion twice, taking the top spot in the Algorithm competition known as "DM i Programmering"

Oxford Computer Science Competition, 1st, University of Oxford, 2013.
For my Numberlink solving software, giving the first fixed parameter polynomial algorithm for the problem.

Demyship, Magdalen College, 2010, 2011.
A historic scholarship awarded to the top students each year.

Les Trophées du Libre, 1st, CETRIL, Soissons, 2009.
First place in the Hobbies category, for my work on the PyChess free software chess suite.

Selected Publications

"Thermodynamic Computing System for AI Applications". By D Melanson, M Abu Khater, M Aifer, K Donatella, M H Gordon, Thomas Dybdahl Ahle, G E

Crooks, A J Martinez, F Sbahi, P J Coles at Nature Communications (NATCOM) 2025.

"Thermodynamic Matrix Exponentials and Thermodynamic Parallelism". By S Duffield, M Aifer, G E Crooks, Thomas Dybdahl Ahle, P J Coles at Physical Review Research (PRR) 2025.

"Thermodynamic Linear Algebra". By M Aifer, K Donatella, M H Gordon, S Duffield, Thomas Dybdahl Ahle, D Simpson, G E Crooks, P J Coles at npj Unconventional Computing (UNCONV) 2024.

"Clustering the Sketch: Dynamic Compression for Embedding Tables". By H Tsang, Thomas Dybdahl Ahle at Advances in Neural Information Processing Systems (NeurIPS) (NEURIPS) 2023.

"Tiling with Squares and Packing Dominos in Polynomial Time". By A Aamand, M Abrahamsen, Thomas Dybdahl Ahle, P Rasmussen at Symposium on Computational Geometry (SOCG) 2022.

"Sharp and Simple Bounds for the raw Moments of the Binomial and Poisson Distributions". By Thomas Dybdahl Ahle at Statistics & Probability Letters (STAT. PROB. LETT.) 2022.

"Similarity Search with Tensor Core Units". By Thomas Dybdahl Ahle, F Silvestri at International Conference on Similarity Search and Applications (SISAP) 2020.

"On the Problem of p_1^{-1} in Locality-Sensitive Hashing". By Thomas Dybdahl Ahle at International Conference on Similarity Search and Applications (SISAP) 2020.

"Subsets and Supermajorities: Optimal Hashing-based Set Similarity Search". By Thomas Dybdahl Ahle, J Knudsen at IEEE Symposium on Foundations of Computer Science (FOCS) 2020.

"Oblivious Sketching of High-Degree Polynomial Kernels". By Thomas Dybdahl Ahle, M Kapralov, J Knudsen, R Pagh, A Velingker, D Woodruff, A Zandieh at ACM-SIAM Symposium on Discrete Algorithms (SODA) 2020.

"Optimal Las Vegas Locality Sensitive Data Structures". By Thomas Dybdahl Ahle at IEEE Symposium on Foundations of Computer Science (FOCS) 2017.

"Parameter-free Locality-Sensitive Hashing for Spherical Range Reporting". By Thomas Dybdahl Ahle, M Aumüller, R Pagh at ACM-SIAM Symposium on Discrete Algorithms (SODA) 2017.

"On the Complexity of Inner Product Similarity Join". By Thomas Dybdahl Ahle, R Pagh, I Razenshteyn, F Silvestri at ACM Symposium on Principles of Database Systems (PODS) 2016.

Media

Rob Aslett. "Si2 Welcomes Normal Computing's Thomas Ahle to Si2 Board", Si2, Aug 2026. Announcement of my election to the Board of Directors of the Silicon Integration Initiative for 2026–2027.

Brian Bailey. "An AI Model Fit For Purpose", Semiconductor Engineering, Jul 2026. Quoted on why compact model representations generalize better.

Tim Scarfe. "The Thermodynamic AI Chip", Machine Learning Street Talk, Jun 2026. Podcast interview on using swarms of AI agents to design and formally verify chips, and on Normal Computing's thermodynamic computing work.

Brian Bailey. "Can AI Create Missing Models?", Semiconductor Engineering, Jun 2026. Quoted on training surrogate models versus steering real systems end-to-end with AI.

Thomas Dybdahl Ahle. "Building an Open-Source Verilog Simulator with AI: 580K Lines in 43 Days", Normal Computing Blog, Mar 2026. My write-up of how we used AI agents to build an open-source Verilog simulation and formal verification stack on top of CIRCT in 43 days.

Brian Bailey. "Multi-Modal AI In EDA Development Flows", Semiconductor Engineering, Jul 2025. Interview on how our company, Normal Computing, is using AI to improve chip design and development.

Jon Lund. "En ulv i fåreklæder", Prosa, May 2021. I was interviewed by the IT workers union on Google's FLoC system and the use of SimHash.

"The Stibo-Foundation supports IT-talents", Stibo, Aug 2016. The announcement of my winning the Stibo Travel grant.

Erik Bärwaldt. "Checkmate: Linux chess program PyChess", Linux Magazine, Jun 2016. Article about my PyChess chess software.

Bidwell, Jonni. "Python: Sunfish chess engine", Linux Format, Jan 2016. Article about my Sunfish chess software.

"The National Team at the Programming World Cup", Computerworld, Jun 2015. Coverage of my team's participation in the ICPC World Finals.

Elkær, Mads. "Denmark's Three Greatest Programmers", Computerworld, Oct 2013. Interview with my team after we won the Danish university programming championship against 34 other teams.

"Winners of the Oxford Computer Science Programming Competition Announced", University of Oxford, Mar 2013. Announcement of my win in the Oxford programming competition with a Numberlink puzzle solver.

"PyChess: Thomas Dybdahl Ahle, Trophées du Logiciel Libre", Les Trophées du Libre, Jul 2009. Video interview about PyChess at the 2009 Trophées du Libre in Soissons, where PyChess won first place in the Hobbies category.

Open Source Projects

Project Owner at PyChess, 2006 - current.
I developed this chess engine and client for Linux desktop, which became the most popular way to play chess on the Free Internet Chess Server. Through the years I have led a team of 4–8 developers and designers, as well as numerous other contributors — including the volunteers who translated it into more than 35 languages. In 2009 we won Les Trophées du Libre in Soissons.

Project Owner at Sunfish, 2012 - current.
A 111-line Python chess engine, which nevertheless has a 2000+ rating on the online Lichess server. Because of the simplicity and focus on teaching good AI techniques, it has become a popular project on GitHub with 2400+ stars and nearly 500 forks. Sunfish was referenced in multiple early applications of neural networks to chess.

Contact

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DBLP List of papers

Google Scholar List of Papers

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